Lab # 02

Designing of Relational Database using MS Visio

Objective:

- Installation and use of UML modeling software (MS Visio).
- To practice ERD diagram on Visio.

Software Tools:

- Microsoft Visio

Theory:

An **Entity Relationship Diagram (ERD)** shows the relationships of entity sets stored in a database. An entity in this context is an object, a component of data. An entity set is a collection of similar entities. These entities can have attributes that define its properties.

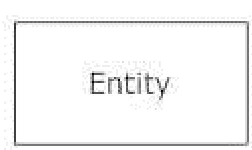
By defining the entities, their attributes, and showing the relationships between them, an ER diagram illustrates the logical structure of databases.

ER diagrams are used to sketch out the design of a database.

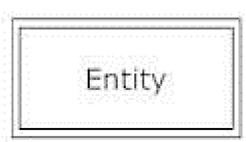
Common Entity Relationship Diagram Symbols

An ER diagram is a means of visualizing how the information a system produces is related. There are five main components of an ERD:

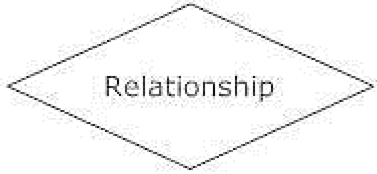
- **Entities**, which are represented by rectangles. An entity is an object or concept about which you want to store information.



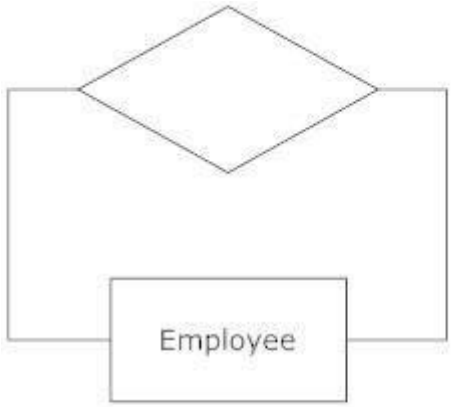
A weak entity is an entity that must defined by a foreign key relationship with another entity as it cannot be uniquely identified by its own attributes alone.



- **Actions**, which are represented by diamond shapes, show how two entities share information in the database.



In some cases, entities can be self-linked. For example, employees can supervise other employees.



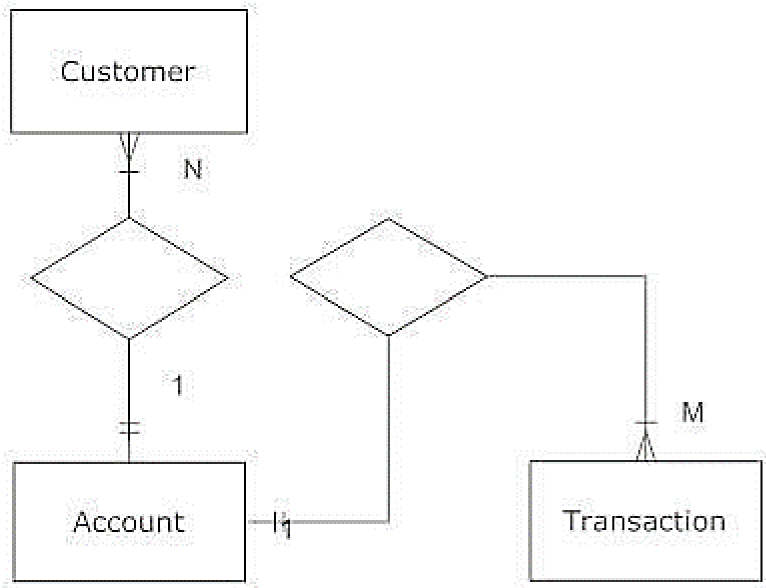
- **Attributes**, which are represented by ovals. A key attribute is the unique, distinguishing characteristic of the entity. For example, an employee's social security number might be the employee's key attribute.

A multivalued attribute can have more than one value. For example, an employee entity can have multiple skill values.

A derived attribute is based on another attribute. For example, an employee's monthly salary is based on the employee's annual salary.

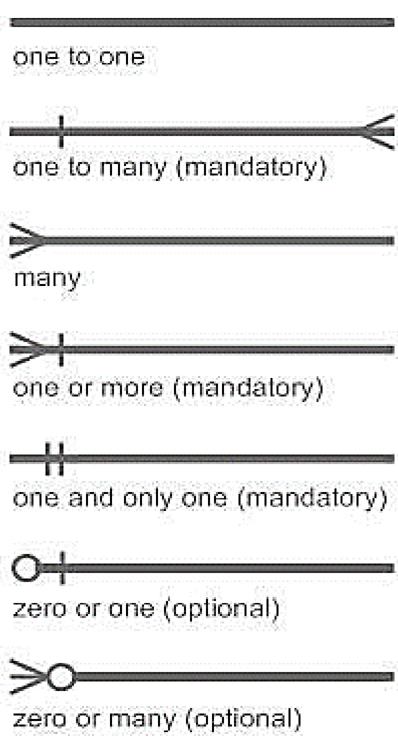
- **Connecting lines**, solid lines that connect attributes to show the relationships of entities in the diagram.
- **Cardinality** specifies how many instances of an entity relate to one instance of another entity. Ordinality is also closely linked to cardinality. While cardinality

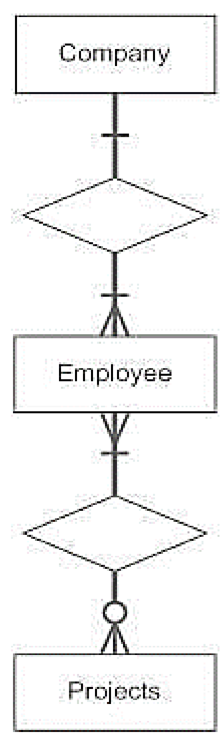
specifies the occurrences of a relationship, ordinality describes the relationship as either mandatory or optional. In other words, cardinality specifies the maximum number of relationships and ordinality specifies the absolute minimum number of relationships.

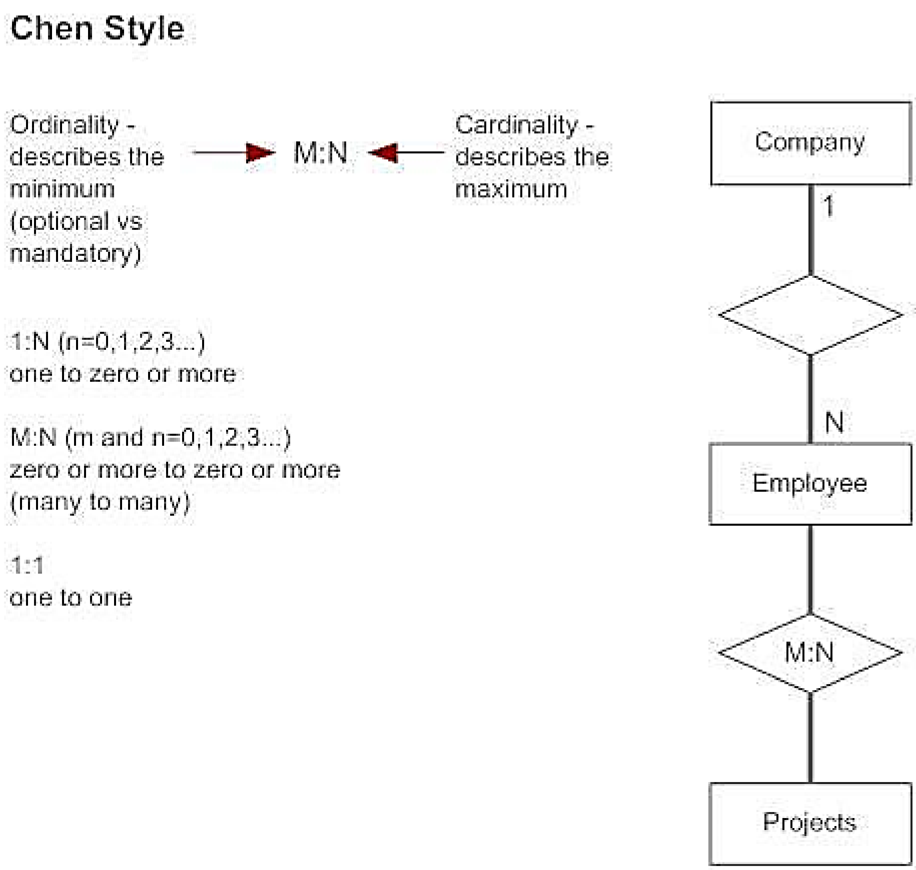


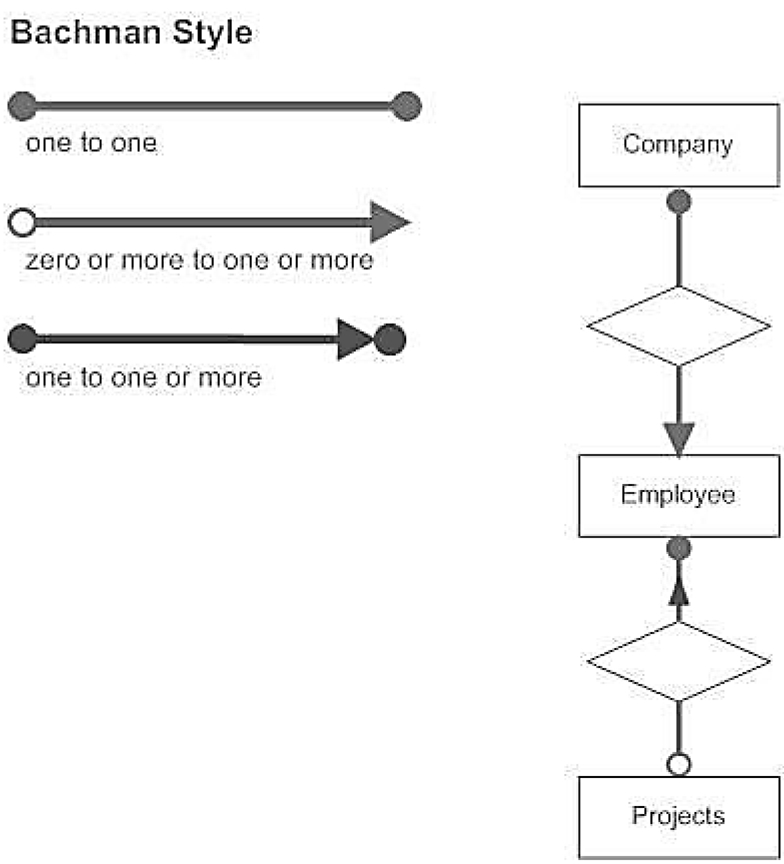
There are many notation styles that express cardinality.

Information Engineering Style






Chen Style
Ordinality - describes the minimum (optional vs mandatory)
M:N
Cardinality - describes the maximum
1:N (n=0,1,2,3...)
one to zero or more
M:N (m and n=0,1,2,3...)
zero or more to zero or more (many to many)
1:1
one to one
Company
1
N
Employee
M:N
Projects


Bachman Style
one to one
zero or more to one or more
one to one or more
Company
Employee
Projects

Martin Style

1 - one, and only one (mandatory)

* - many (zero or more - optional)

1...* - one or more (mandatory)

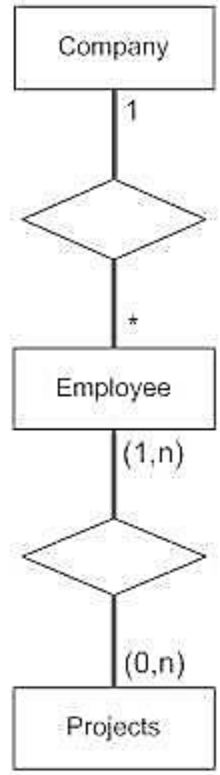
0...1 - zero or one (optional)

(0,1) - zero or one (optional)

(1,n) - one or more (mandatory)

(0,n) - zero or more (optional)

(1,1) - one and only one (mandatory)



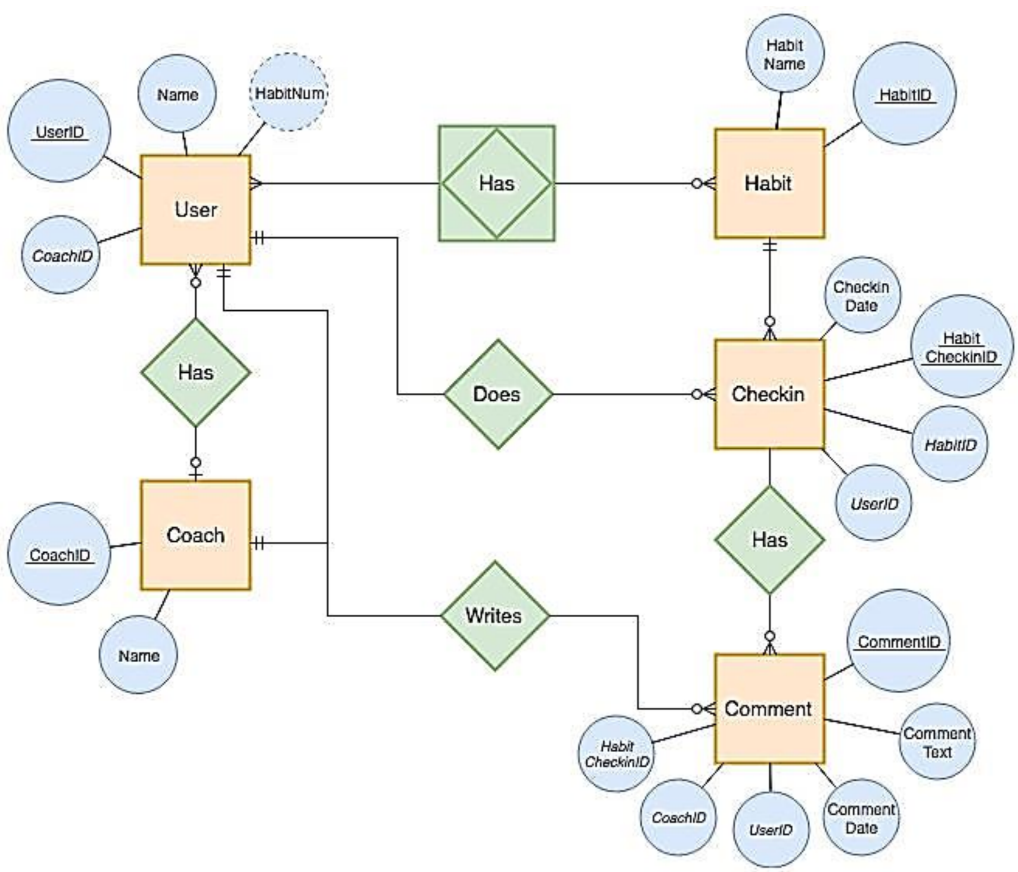


Figure 1: ERD Diagram

Microsoft Visio is software for drawing a variety of diagrams. These include flowcharts, org charts, building plans, floor plans, data flow diagrams, process flow diagrams, business process modeling, swimlane diagrams, 3D maps, and many more. It's a Microsoft product, sold as an addition to MS Office. Visio 2016, the latest version, comes in three editions: Visio Standard, Visio Professional, and Visio Pro for Office 365

Lab Work

Microsoft Visio 2007 installation:

1. Open the MS VISIO 2007 folder and click on setup, Enter your Product key and click *"continue"*

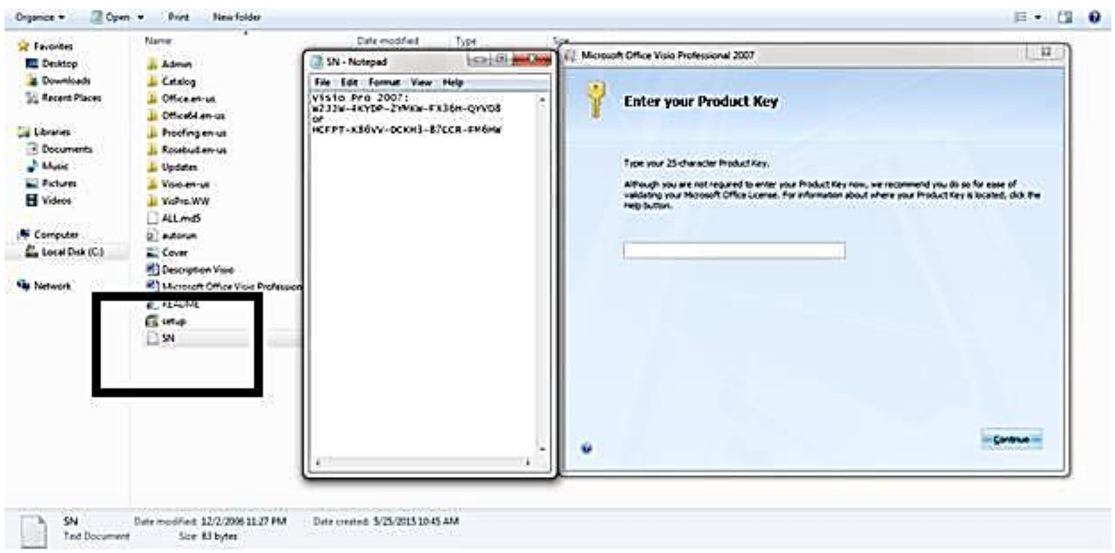


Figure 1: Step 1 Initialize the Visio Step up

Figure 2: Installation Key

2. Tick the Terms and Condition and Click “*Continue”* (see *Figure 3)*

Figure 3: Step 2 Terms and Condition

3. Click on Install now (*see Figure 4 and Figure 5*)

Figure 4: Step 3 Install Now

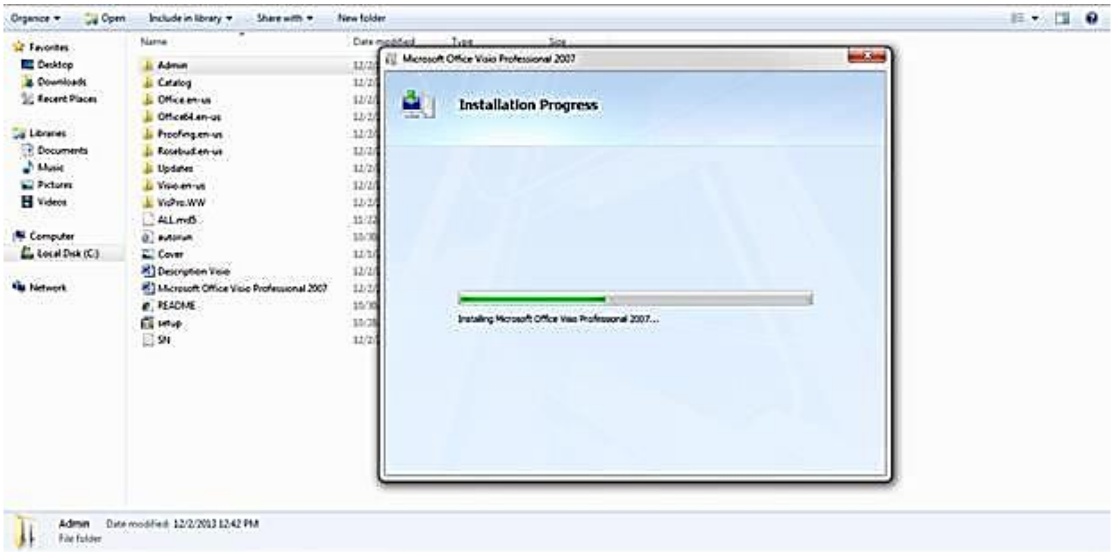


Figure 5: Installation Progress

4. Installation has been completed and Click “Close”

Figure 6: Closing window

How to Use MS Visio

Please follow the step by step process to use the MS VISIO

Step 1: Go to the Start Menu and select Program MS Visio and open a New Visio drawing (*see Figure 7*)

Figure 7: Visio Start

Step 2: Open the file menu and select New then Move your cursor over "Basic Diagram" and select (*see Figure 8*)

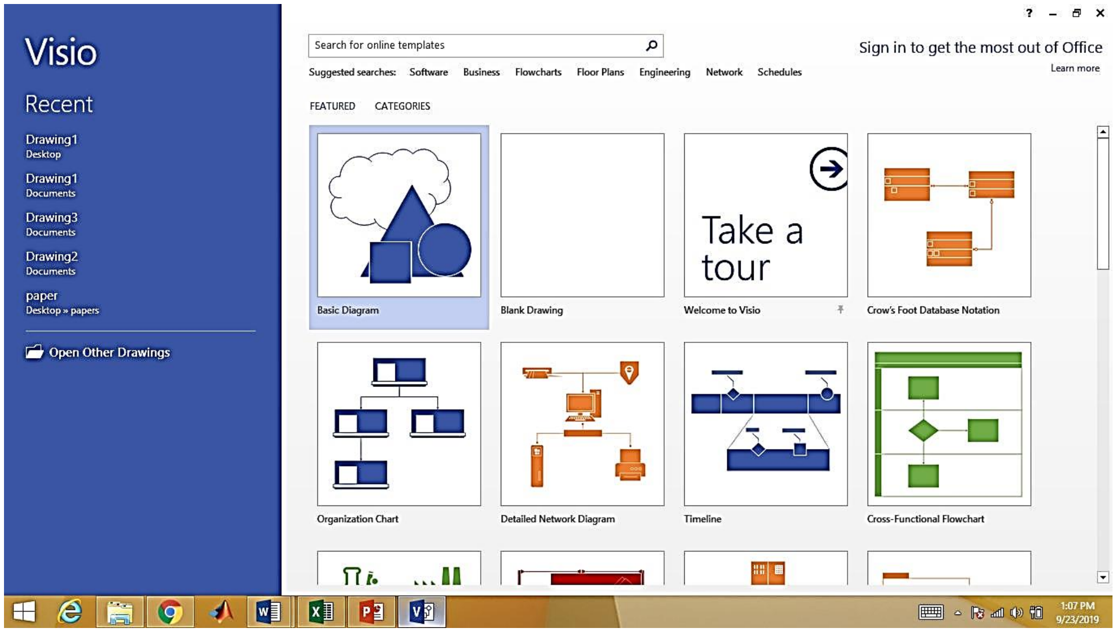
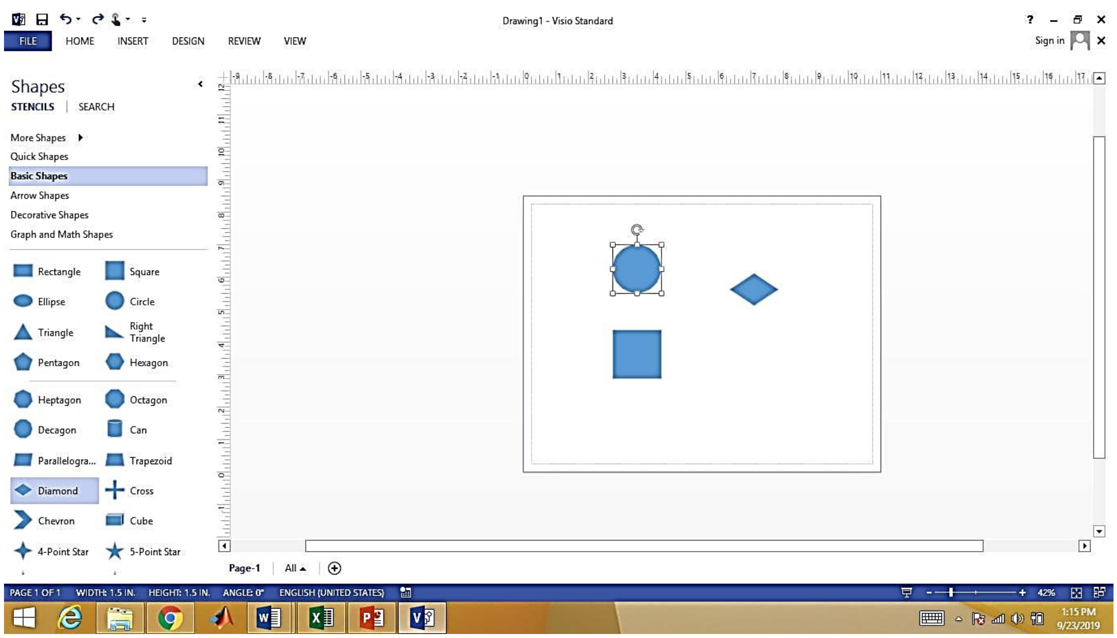
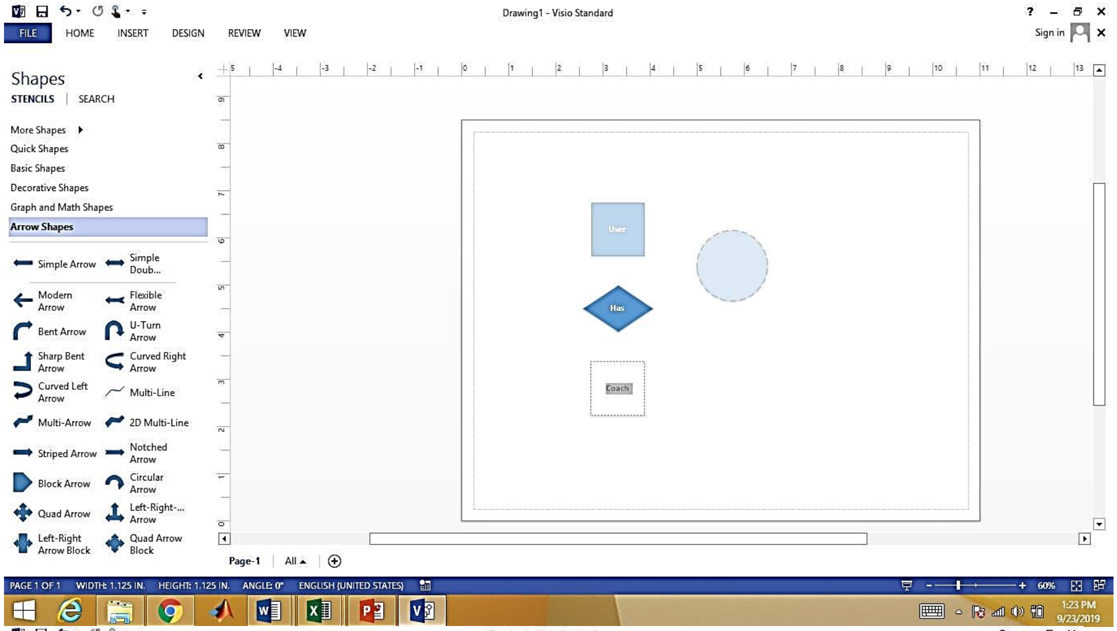


Figure 8: Select Template

Step 3: Select a shape from the Shapes menu and drag it to the workspace.

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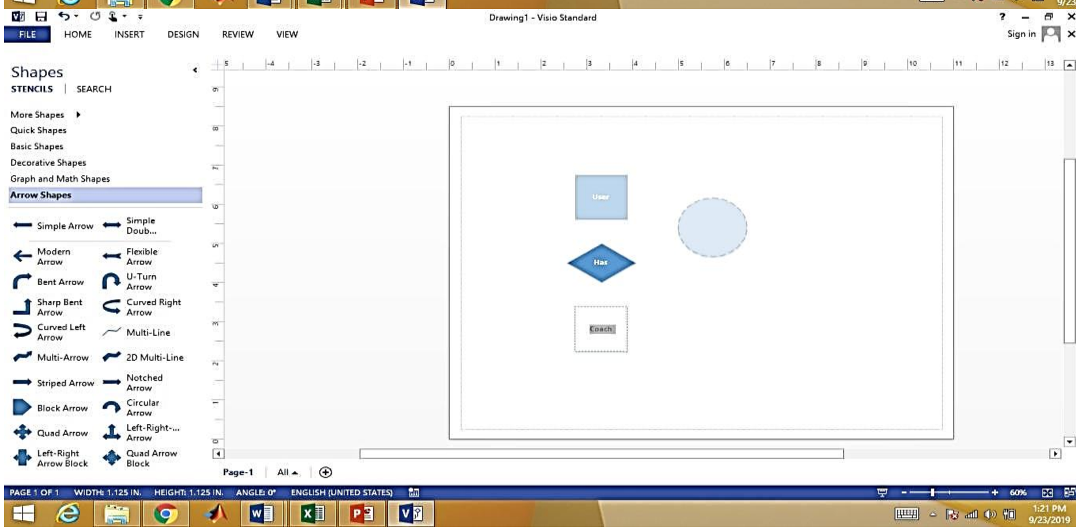
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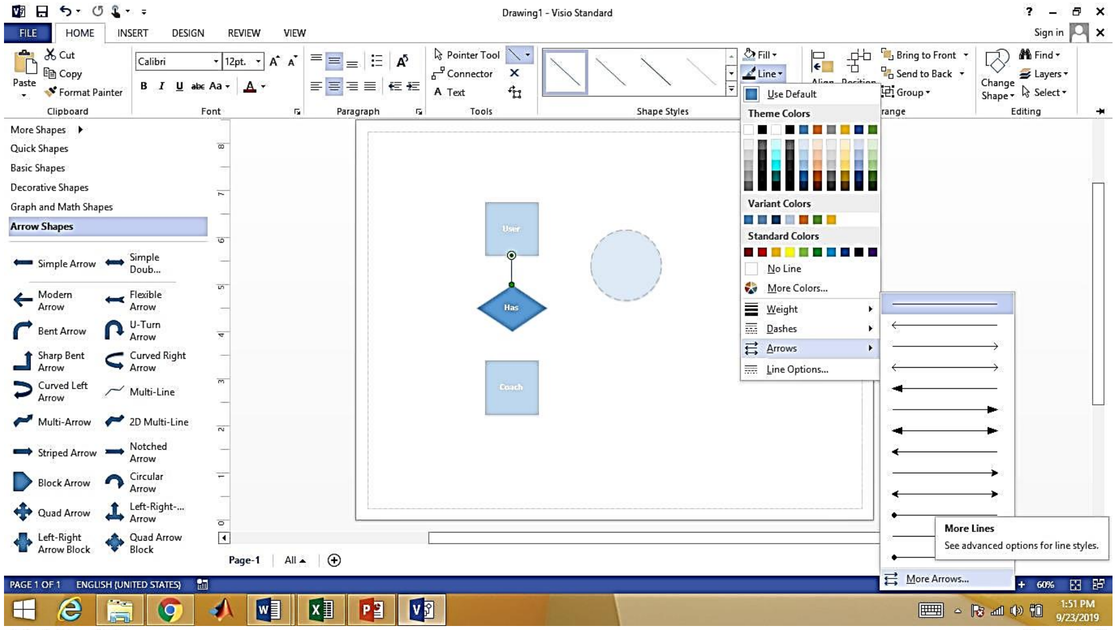
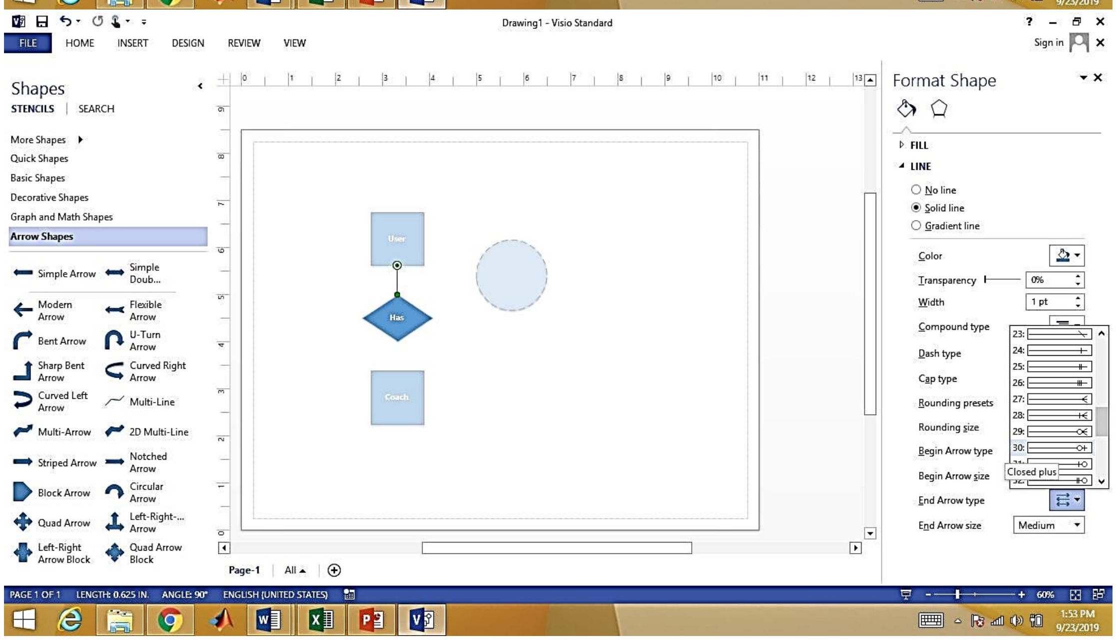
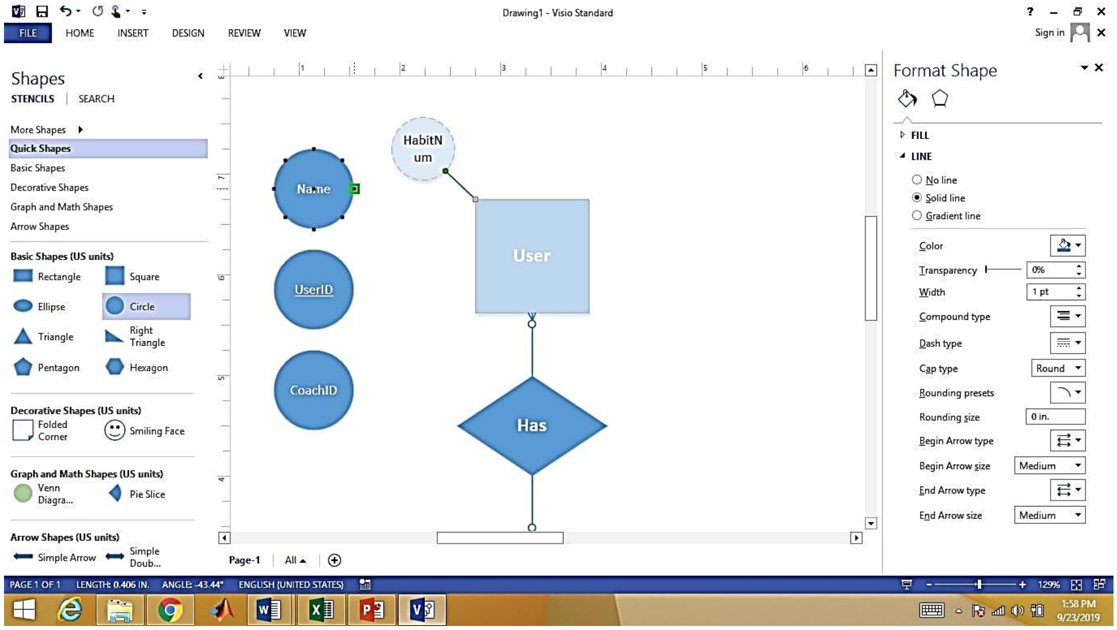




Step 4: On the toolbar, click the connector tool. The connector tool will appear highlighted, and will remain active until it is deselected. Double click on connector and naming the line.



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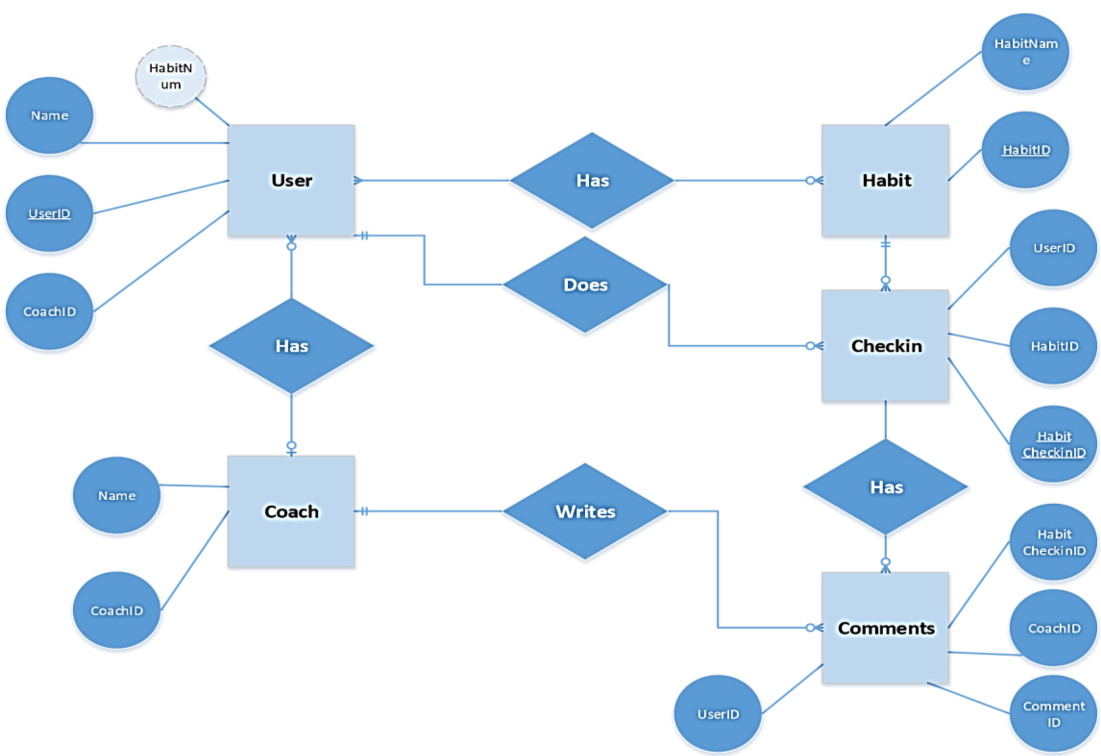

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TASKS:

A. Design ERD for a vehicle enterprise in MS Visio in which a dealer sells vehicles. At workshop, these vehicles must be maintained by mechanics. Mechanics send the charges ship to the dealer. The database should include the following information:

- Basic information about vehicle: Vehicle type, registration number, model.
- Basic dealer information: dealer name, contact number, dealer cnic number.
- Sales information: date, model.
- Mechanic information: mechanic name, cnic number, address

B. Draw this ERD in Visio